OMNIKON NATIONAL HACKATHON 2026

VERIFRAME

Verify what you see. Trust what you share.

PROBLEM Omni_CyberTech_4 — Detecting Deepfake and Manipulated Media

THEME Cybersecurity, Blockchain & Digital Trust

Team codeX

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Shipped this phase: a real forensic detection engine, JWT + bcrypt auth, a data layer, a FastAPI model service, Docker, CI, and 20 passing tests.

The trust gap in synthetic media

- **Manipulated media outpaces review** — Face swaps, cloned voice and edited video erode public trust faster than anyone can verify by hand.
- **Reviewers work across scattered tools** — Fact-checkers and newsrooms lack one forensic workspace for images, video and audio.
- **Detectors decay** — New GANs, diffusion models and voice cloners keep appearing; any fixed model goes stale.
- **Provenance rarely reaches humans** — Signals like C2PA exist but are not surfaced where an analyst makes the call.

WHAT ANALYSTS ACTUALLY NEED

Explainable evidence

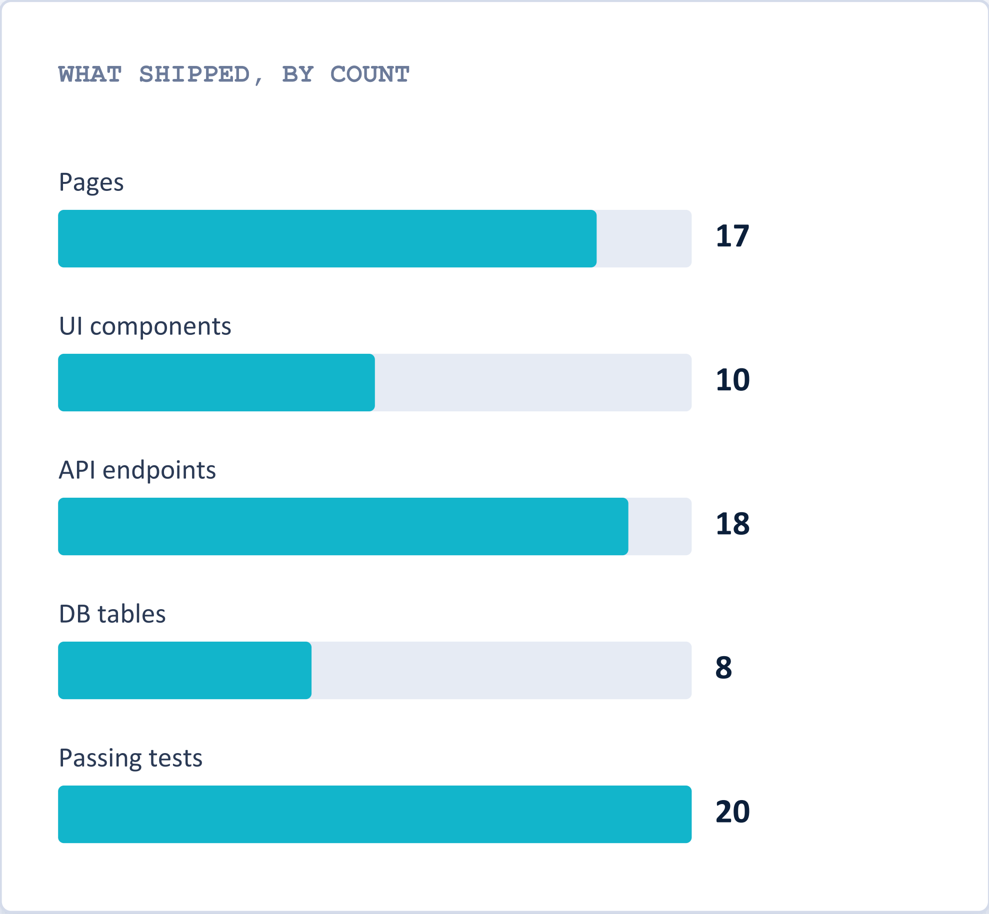
not an opaque score

Heatmap · EXIF
C2PA · audit trail

A human-override path on every verdict

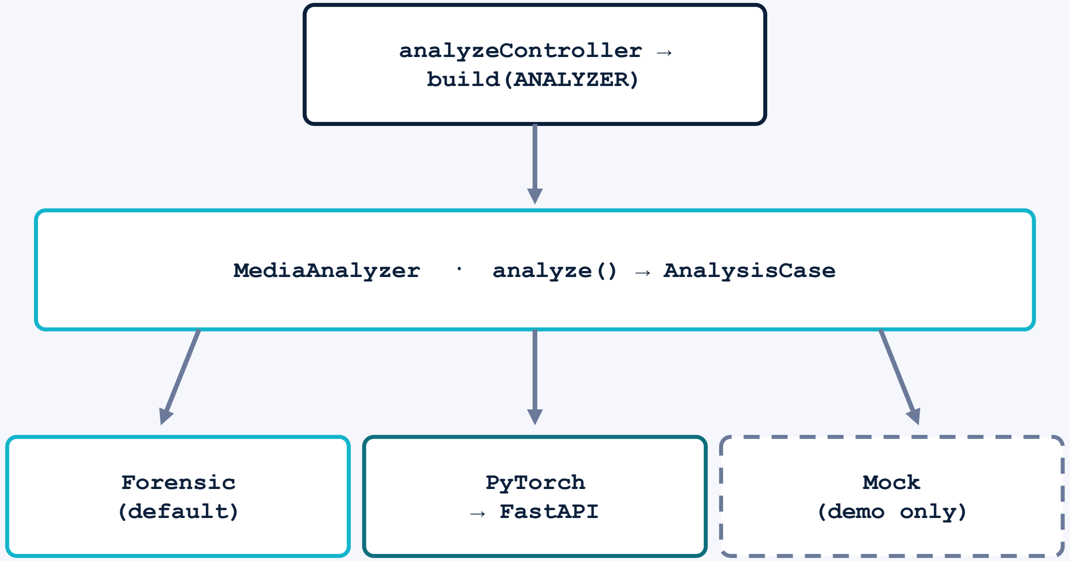
The platform is built and running

- 17-page React 18 + TypeScript + Vite + Tailwind SPA, fully routed
- SOC dashboard shell: Navbar, Sidebar, Footer, recharts analytics
- Express 4 TypeScript REST API — ~18 endpoints across 6 route groups
- Interactive forensic viewers: anomaly heatmap, video timeline, audio waveform
- 8-table PostgreSQL schema wired via db/migrate.ts + a PgStore backend
- Runs with zero external services — local API + in-memory store, no GPU



A real forensic engine behind one seam

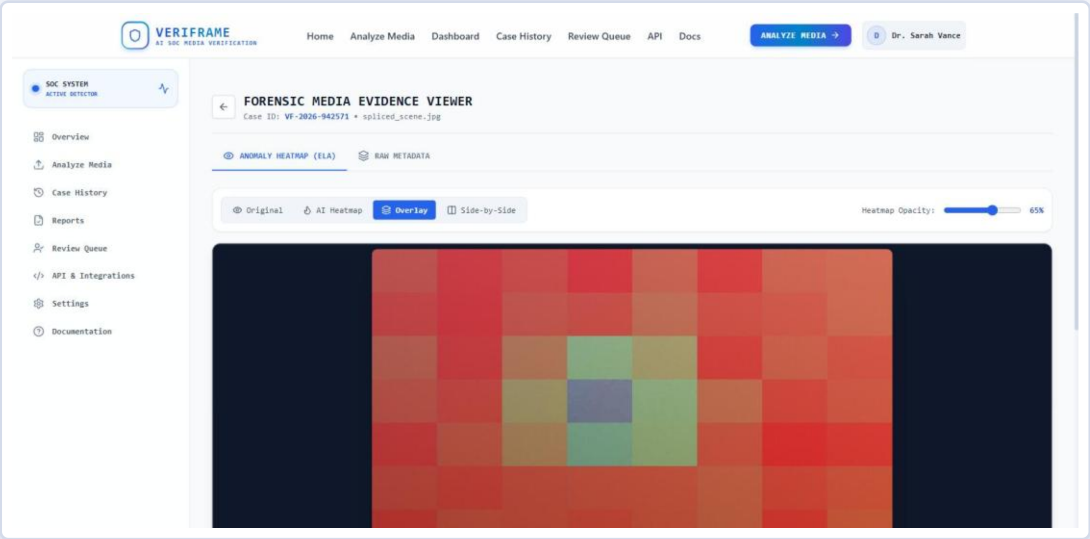
- One MediaAnalyzer interface isolates the whole engine
- analyze(input) → Promise<AnalysisCase> — one env-selected injection point
- Default engine reads bytes: ELA heatmap · noise stats · EXIF · C2PA probe
- Real signal, no ML model — output labelled modelVersion "forensic-v1-noml"
- Weighted aggregation → verdict, authenticity score, manipulation probability
- Switching to the model service is one env flip: ANALYZER=pytorch



Forensic engine computes signal from the file itself — no model is trained, loaded, or called. A trained model is the third plug.

Frontend surface — 17 pages

- Landing, Login, Register, Dashboard, Analyze, Processing, Results
- Evidence viewer, Case History, Case Detail, Provenance, Reports
- Human Review Queue, API Docs, Settings, Help & Learn, 404
- Drag-drop upload or URL input, plus three one-click demo samples
- Recharts dashboard: KPI cards, 7-day trend, verdict-distribution donut
- Heatmap canvas: original / heatmap / overlay / split + opacity slider



Evidence viewer — real 8×8 ELA residual grid over the uploaded file

Heatmap is computed from pixels by sharp — not model gradients. Video/audio viewers are real UI, single anomaly marker pending the model service.

05 · FEATURES IMPLEMENTED

API, auth & working end-to-end flows

- ~18 endpoints: auth · analyze · cases · reviews · reports · stats · keys · health
- helmet · CORS origin allowlist · global + per-route rate limits · zod on every body
- Multer in-memory uploads capped at MAX_UPLOAD_MB (default 50) + mime allowlist
- analyze and every mutation require a Bearer token; read paths use optional auth
- Store interface: MemoryStore by default, PgStore when DATABASE_URL is set

Stack & data layer

React SPA — 17 pages, server-authoritative

Vite 5 dev :3000 · Bearer token · slim offline fallback on read views only

Express REST API :5000

helmet · rate-limit · zod · jsonwebtoken · bcryptjs · x-powered-by off

Forensic engine

sharp (ELA) + exifr (EXIF) + C2PA probe · no PyTorch / ONNX in Node

Store interface

MemoryStore (default, resets) | PgStore when DATABASE_URL set

PostgreSQL · schema.sql + migrations

applied by db/migrate.ts, backs PgStore · 8 tables

FastAPI ai-engine :8000

freq-heuristic-v1 today · trained torch/timm backend = MODEL_WEIGHTS drop-in

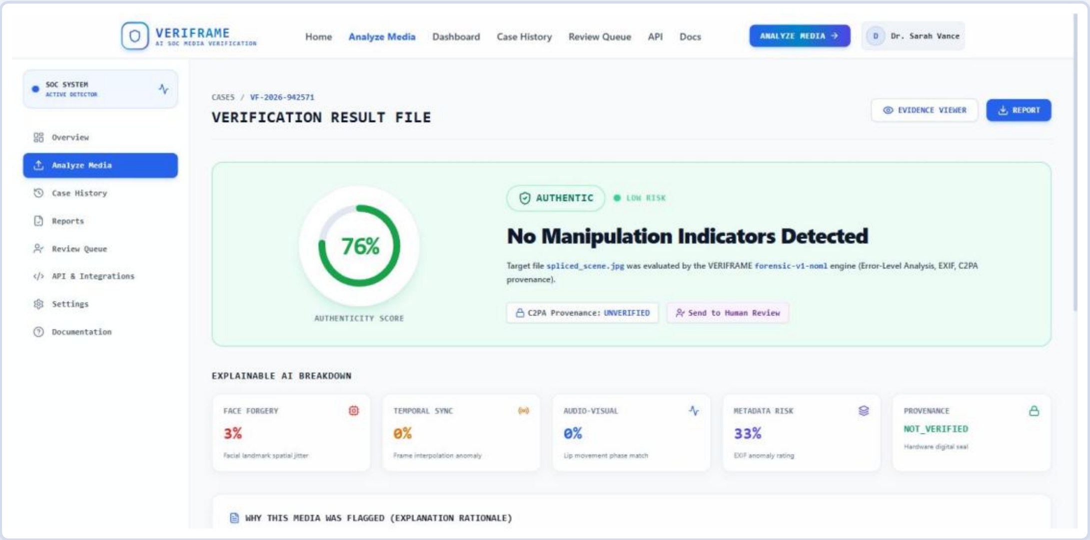
READ THIS AS

The product surface, forensic engine and data layer are real.

Still genuinely absent: ffmpeg, and any trained model weights.

The model seam & inference service

- services/analysisEngine/index.ts build() switches on ANALYZER
- ForensicMediaAnalyzer default — ELA, noise stats, EXIF, C2PA byte-marker probe
- C2PA: optional c2pa-node does crypto verify; default is a JUMBF marker scan
- PyTorchMediaAnalyzer built — POSTs to FastAPI, falls back to forensic on failure
- FastAPI ai-engine runs today: freq-heuristic-v1, a 2-D power-spectrum check
- Trained torch/timm backend is a MODEL_WEIGHTS drop-in — no weights in repo



Live forensic verdict — engine tag "forensic-v1-noml", C2PA UNVERIFIED

Every score on this page traces to a computation on the uploaded file.

Build challenges & deliberate trade-offs

Scope	Scoping 17 SOC-grade pages + bespoke forensic viewers in hackathon time
Contract	One analyzer contract covering image, video & audio before any model existed
No-ML engine	Writing a real forensic engine that fills the same contract a model will
Client	Making the client server-authoritative while read views stay resilient offline
Deferred	Trained face-forgery / voice models + ffmpeg frame & audio extraction

The core problem: defining the AnalysisCase contract for all three modalities before a model existed, then writing a real engine to fill it.

Domain risk we design around

Model staleness

New GANs & voice cloners outpace any fixed detector

MITIGATION Continual retraining loop from reviewer overrides

Compute cost

Real-time video inference is GPU-heavy vs. latency goals

MITIGATION Lightweight edge triage pass before deep model

False positives

Flagging authentic media is reputationally costly

MITIGATION Ensemble + provenance; review-required thresholds

Language bias

Lip-sync & synthetic-voice cues carry dialect bias

MITIGATION Human-in-the-loop queue for borderline calls

Heuristic recall

Forensic heuristic, not ML — bounded on high-quality fakes

MITIGATION Trained model behind the same seam (Phase 3)

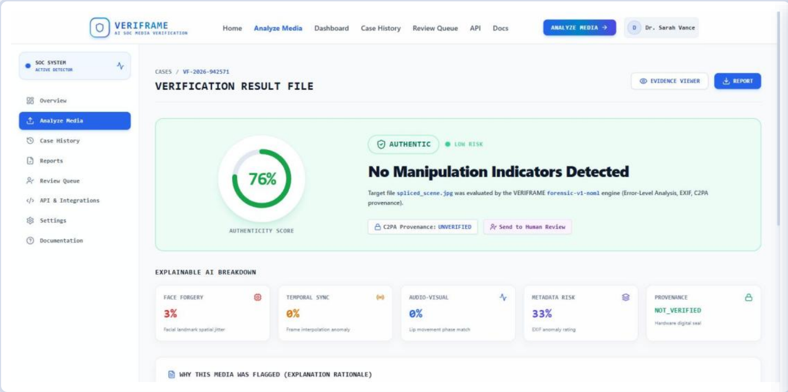
Fill the model slot — the seam is already wired

1. Train & serve real model weights behind the existing FastAPI service
2. Face-forgery CNNs + synthetic-voice models; keep the forensic fallback
3. Bundle c2pa-node for default crypto C2PA; add ffmpeg frame & audio extraction
4. Model-gradient heatmaps + real per-frame / per-timestamp anomaly scores
5. Persist users & API keys in Postgres — bcrypt + JWT are already live
6. Feed Human Review Queue overrides into a continual retraining dataset

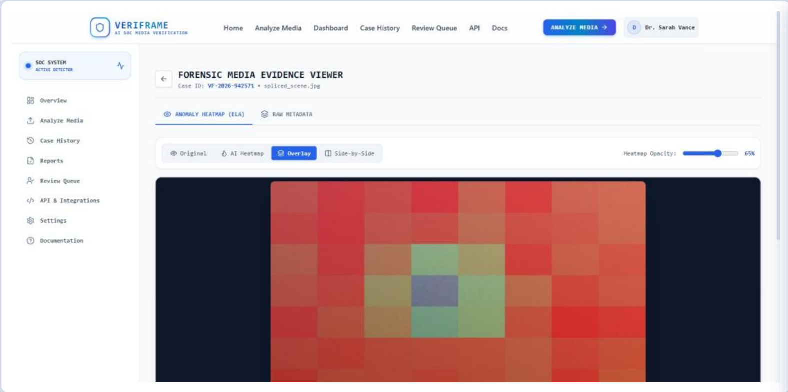
Every handoff point already exists: FastAPI running, PyTorch client built, ANALYZER=pytorch flip ready.

11 · SCREENSHOTS & LIVE DEMO

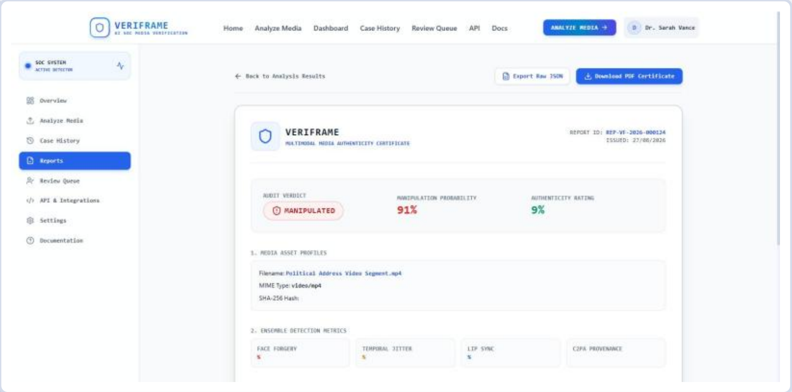
A fresh upload returns a real verdict



Results — real forensic verdict + breakdown



Evidence — real 8×8 ELA heatmap



Report — export JSON / print PDF

- Demo runs on the local Express API — ANALYZER=forensic, in-memory store, no GPU
- Landing → Analyze → 7-stage processing → Results verdict banner
- A fresh file upload returns a real score from ELA, noise stats, EXIF, C2PA probe
- Provenance panel is driven by a real C2PA probe + EXIF on the uploaded file
- Review queue + Reports: verdict override, export JSON, print PDF
- The three 1-click samples + dashboard KPIs are fixtures — flagged on screen

The platform, the forensic engine and the model seam all work today.

- Default analyzer is real forensic signal — ELA, noise, EXIF, C2PA — no ML.
- Trained model is the last drop-in; nothing above the seam changes.
- Handoff is live-wired: FastAPI running, PyTorch client built, env flip ready.
- Hardened: JWT/bcrypt auth, rate limiting, data layer, Docker, CI, 20 tests.
- No accuracy figures claimed — benchmarking on FF++ / DFDC / Celeb-DF is Phase 3.